

High resolution **von Hamos X-ray Spectrometer** using **HAPG for extended sources in a broad energy** **range: VOXES**

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Abstract

The VOXES project's goal is to realise the first prototype of a high resolution and high precision von Hamos X-ray Spectrometer for a broad energy range, using Highly Annealed Pyrolytic Graphite (HAPG) crystals combined with Charged Coupled Devices (CCDs) and/or Silicon Drift Detectors (SDDs). The aim is to deliver a cost effective system having an energy resolution at the level of eV for X rays energies from about 2 keV up to tens of keV, able to perform sub-eV precision measurements. The performances of VOXES will approach and overtake those of the newly developed Transition Edge Sensors (TES), which achieve eV resolution measurements for X rays of few keV, with the drawback that TES have high costs, extremely reduced active area, rather long recovery time and are difficult to be handled, due to the massive cryogenic system needed. VOXES spectrometer will have a reduced cost, a bigger active area, higher efficiency and will be easier to handle. For the first time, moreover, the system will work as well with extended X rays sources (i.e. not punctiform, nor collimated). In order to realise the proposed system we shall collaborate with a world leading firm developing HAPG crystals (OPTIGRAPH [1]), such as to optimise their structure and geometry for VOXES. We will rely on the principal investigator (PI) and his team's worldwide recognised expertise in realising and operating cutting edge technologies for high precision X-ray detection systems, achieved in performing experiments at the DAΦNE collider, at PSI and KEK laboratories, and at the LNGS-INFN. The VOXES system will be qualified by performing real measurements of exotic atoms at the PSI laboratory and/or DAΦNE and/or J-PARC laboratories. There are many applications of the proposed spectrometer, going from fundamental physics (precision measurements of exotic atoms at DAΦNE collider and J-PARC; precision measurement of the K^- mass, solving the existing puzzle; quantum mechanics tests) to synchrotron radiation and X-FEL applications, astronomy, medicine and industry.

1) Background of the proponent(s)

The PI, as well as the team which he's coordinating for realising the VOXES project, are experts in developing and using high precision X-ray detector systems, based on SDDs and CCDs. The PI's expertise was gained firstly during his PhD fellowship, in the framework of the SIDDHARTA collaboration, to which (see CV) he gave important contributions. He performed the first measurement ever of the kaonic helium 3, object of his PhD thesis, as well as of other exotic atoms measurements. The PI then continued his research activity both in the framework of the SIDDHARTA collaboration, with assigned responsibilities, and within the VIP collaboration at the LNGS-INFN, where a system based on CCD detectors was used to test the Pauli exclusion principle by performing atomic transitions measurements.

The expertise of the PI and of the participating and collaborating institutions: LNF-INFN, SMI-Vienna, Milan Polytechnic, which are worldwide recognised for realising cutting edge technologies performing spectroscopic X-ray measurements, together with that of the OPTIGRAPH company, one of the leading companies in the design and realisation of the HAPG crystals, is the best guarantee for the successful realisation of the VOXES project.

2) State of art

The possibility to perform very high resolution measurements of the X rays emitted in various processes is becoming a priority in many fields of fundamental science, from particle and nuclear physics to quantum mechanics, as well as in astronomy and in many applications using the synchrotron light sources or X-FEL beams, in biology, medicine and industry.

The X-ray detection systems are presently passing through a remarkable fast development process, with new type of detectors being constantly realised, having steadily improved performances in terms of efficiency, energy resolution and costs.

In spite of the fast evolving radiation detectors market, the aim of having accessible large area X-ray detector systems, able to achieve eV energy resolution for energies going from few keV up to tens of keV, and capable to measure not only well-collimated sources, but also the extended ones is still to be reached.

Within VOXES we plan to realise the first prototype of a system, capable to measure with eV energy resolution X rays from 2-3 keV up to tens of keV, which works with extended sources.

In what follows we give a short overview of the state of art of spectroscopic solid state X-ray detector systems and of the HAPG crystals, and an example of the need to have better X-ray energy resolution detectors in fundamental science.

High resolution spectroscopic X-ray detectors

Silicon and, more generally, solid state detectors, such as Silicon Drift Detectors (SDDs) or Charged Coupled Devices (CCDs), have been intensively used as position and energy sensitive detectors. Recently, their use as large area spectroscopic detectors has been explored in the framework of the DEAR and SIDDHARTA experiments [2,3,4,5,6], with lot of success. However, when used for precision transition lines widths measurements these devices are limited by their **intrinsic resolution** related to the Fano Factor [7, 8, 9] and to the electronic noises [10], which were investigated in detail. Such detectors have resolutions at 6 keV energy of about 120 eV (FWHM) at best. In the past, experiments performed at the PSI laboratory measuring pionic atoms [11] pioneered the possibility to combine these type of detectors (CCDs) with crystals, to enhance the energy resolution. The used crystals were of silicon type and the energy range achievable to the system was limited to few keV, due to the crystal structure. Moreover, the capacity of these systems to work with extended sources was rather limited, due to the rigidity of the crystals

Other type of high resolution detectors being presently under development are the Transition Edge Sensors (TES). The achievable energy resolution is excellent (few eV at 6 keV), but the active area is still very small [12]. Moreover, the cost of these detectors is prohibitively high and their use rather laborious, due to the large complex cryogenic system needed to reach the operational temperature of ~ 1 K.

In these conditions, a new methodology to overcome these problems comes from the development, in the last decade, of the Highly Annealed Pyrolytic Graphite crystals (HAPG,[1, 13]), which were proven to be able to perform measurements of X rays from few keV up to tens of keV energy with energy resolutions of $\Delta E/E < 10^{-3}$, when combined with standard CCD detectors. Thanks to the intrinsic properties of these crystals, like **high reflectivity, efficiency and spatial resolution**, they proved to be an ideal solution for focusing soft X rays on position sensing detectors [14] ; in addition, their high flexibility allows to build them with ad-hoc geometries leading to the possibility to reach, through the optimisation of their curvature radius, length, thickness and mosaicity, **high collection angles and efficiencies**. In what follows more details of these type of crystals are given – since they are going to be an important part of VOXES project.

Focusing crystals for X rays: HAPG

Highly Annealed Pyrolytic Graphite (HAPG) is a mosaic crystal, consisting in a large number of nearly perfect small crystallites. The unique structure of HAPG crystals enables them to be highly efficient in diffraction in an energy range between 2 keV and several tens of keV. Furthermore, thin HAPG crystal films can be **easily bent**, allowing the production of ad-hoc optimised shapes. The diffraction properties influencing the energy resolution of HAPG films are determined by the mosaicity and the intrinsic width of Bragg reflection. Mosaicity makes it possible that even for a fixed incidence angle on the crystal surface, an energetic distribution of photons can be reflected, because each photon of this energetic distribution can find a crystallite plane at the right Bragg angle (see Fig. 1).

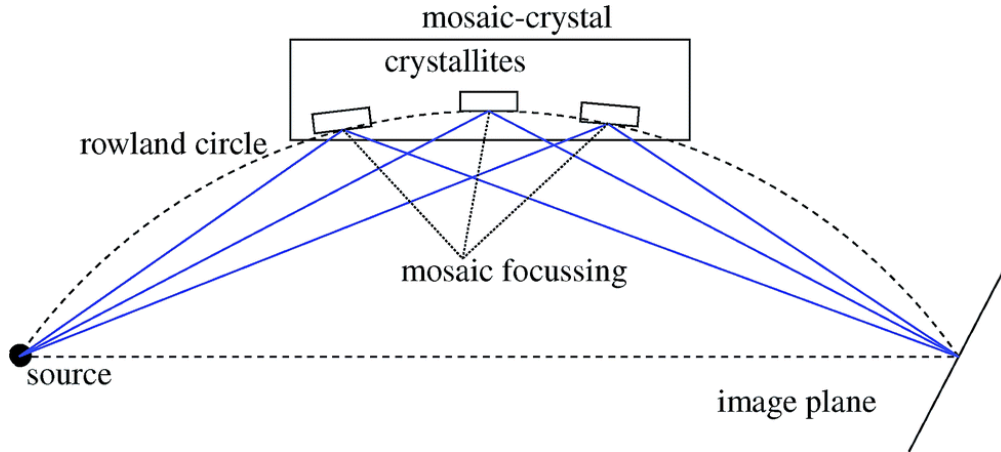


Figure 1: Mosaic focusing for a flat mosaic crystal; due to the mosaic structure, there are crystallites appropriately oriented and positioned [15]

The width of the energetic distribution depends on the mosaic spread. The mosaicity is also responsible for the **dramatic increase of the integrated reflectivity** in comparison to perfect crystals in an energy range between 2 keV and several tens of keV [5].

The idea to **combine the dispersion of a flat crystal with the focusing properties of cylindrically bent crystals** was put forward from von Hamos in 1933, in the so called von Hamos spectrometer [16], and has already proven its high-resolution capabilities [17]. The principal elements of such a spectrometer are the X rays source, in this case defined by a rectangular slit, a cylindrically bent crystal, and a position sensitive detector located on the crystal axis of curvature (see Fig. 2).

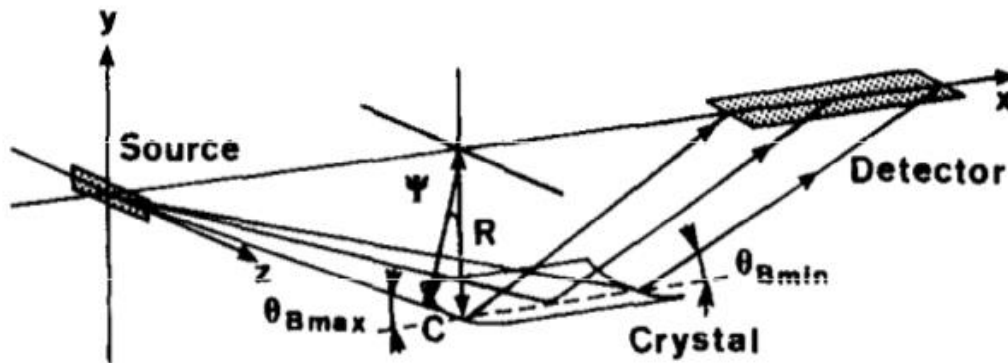


Figure 2: Schematic drawing of the von Hamos spectrometer geometrical principle of operation. The crystal is curved around the x-axis. The figure is not to scale [17].

In the von Hamos geometry the crystal is bent around the x-axis which is parallel to the direction of dispersion and provides focusing in the non-dispersive z-direction. The reflected X rays form an image (a slightly curved line) on the detector surface. For a fixed position of the components, an incident X ray location on the detector corresponds geometrically to a particular Bragg angle and hence to a particular X ray energy through the Bragg relation:

$$n\lambda = 2d\sin\Theta_B \quad (1)$$

where n is an integer number, d is the interplanar distance of the crystal, λ and Θ_B are the wavelength and Bragg angle, respectively. Such focusing, called vertical focusing, permits at one position of the spectrometer components, data collection over an energy bandwidth limited primarily by the crystal and detector lengths (x-extension). In order to study a greater energy interval the central Bragg angle is adjusted by translation of the crystal and correspondingly of the detector along their axes. The source-to-crystal and crystal-to-detector distances are varied but kept equal one to each other. The position of the crystal center from the slit is given by

$$x_C = R\cot\Theta_B \quad (2)$$

where R is the crystal radius of curvature and Θ_B is the central Bragg angle. The detector distance x_D is twice that of x_C . The major component to the spectrometer resolution loss stems from the extended source size. The energy resolution ΔE loss due to the slit width Δy can be approximated by:

$$\Delta E \sim E\cos^2\Theta_B(\Delta y/2R) \quad (3)$$

where E is the energy, Θ_B the Bragg angle and R the crystal radius of curvature [17]. The von Hamos focusing geometry gives **high collection efficiency**. The ratio of efficiencies between a von Hamos spectrometer and a flat crystal one working at the same geometry is equal to $R\theta/a$ ([14], see Fig. 3) where R is the radius of curvature of the crystal, θ is its angular aperture, and a is the source size. For a small source size this ratio may be as large as 10^2 - 10^3 . Note that the von Hamos geometry has focusing properties and, hence, **high collection efficiency in a full spectral bandwidth $\Delta\lambda$** . In other focusing schemes the wavelength band and the efficiency are mutually exclusive.

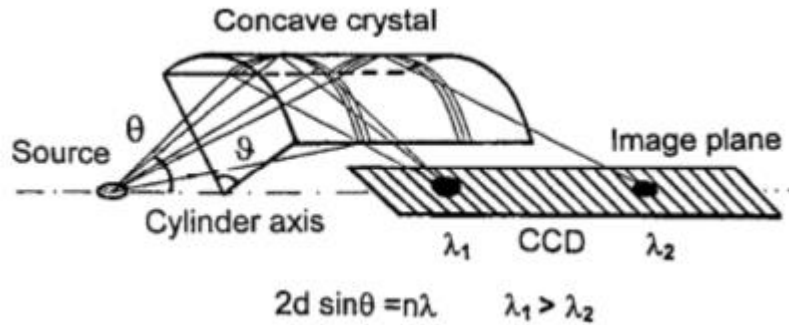


Figure 3: Schematic diagram of the von Hamos spectrometer: the x-ray source and the detector plane lie on the axis of the cylindrical crystal. Each wavelength, after reflection from the crystal arc, is focused to a point lying on the spectrometer axis [14].

The excellent features of von Hamos spectrometers using HAPG crystals have been already tested and proved; several studies have been carried out showing their excellent focusing properties [14] and resolution [17].

However, in all these experiments the crystals have been used to focus **very small and collimated sources**. In the past an attempt to use silicon crystals for wider sources has been carried out [11] where the pionic hydrogen and deuterium X-ray lines have been measured, but these lines are in the very small range of 2.4-3.1 keV and the system had a very small efficiency. Another proposal has also been put forward in 2002 [18], still with non-HAPG crystals, but it ended with the conclusion that also for

efficiency reason the measurement would need a very long data taking period. **The VOXES project will exploit the high collection efficiency of the HAPG crystals and their versatile geometry in order to realise a high resolution X-ray detector for a broad energy range for extended sources.**

Why the need for higher resolution systems (VOXES)? Example from hadronic physics

In what follows we give an example of the need of the VOXES system in hadronic physics. During the last decades various experiments measured exotic atoms (pionic or kaonic ones) transitions, meaning X rays in the range of few keV, in order to extract the contribution coming from the strong interaction between the pion or kaon and the nucleus, and to obtain extremely important information to understand the QCD in non-perturbative regime. Unveiling the QCD's mechanisms in non-perturbative regime, especially in the strangeness sector, will allow to have a closer look to the interplay between spontaneous and explicit chiral symmetry breaking mechanisms, which are supposed to contribute to the mass generation of the utmost part of the normal matter in the Universe. On the strength of the low-energy antikaon-nucleon interaction it depends, as well, whether the (core of) the neutron stars might harbor exotic phases associated with strangeness. The DEAR and SIDDHARTA experiments [2, 3, 4,5,6] at the DAΦNE collider at LNF-INFN and the E570 [19] at KEK showed very clearly how the possibility to measure X rays between 2 and 15 keV with few eV precision represents the only possible way towards a full and comprehensive understanding of the low-energy strong interaction in the strangeness sector. In order to do this, detectors like CCDs, SDDs and “Si(Li) X-ray detectors” were used, leading to very important measurements at the LNF-INFN (DEAR, SIDDHARTA, [6, 5]) and at laboratories like KEK, (E570,[19]) and PSI (pionic hydrogen experiment, [20]).

However, the limited intrinsic resolution (120 eV at 6 keV of FWHM at best) of these detectors denied the possibility to perform precise determination of some fundamental parameters for theoretical models like, for instance, the shifts and the widths induced by the strong interaction of the $3d \rightarrow 2p$ transition lines of the kaonic helium isotopes ($K^3\text{He}$ and $K^4\text{He}$), which are expected to be below 1 eV in some models and around few eV in some other models, with dramatic consequences in the second case. The same is true for another extremely important measurement: the negatively charged kaon mass, which can only be extracted by a very high precision kaonic atoms transitions measurement; this measurement has an important impact in particle and nuclear physics. Moreover, the possibility to measure precisely the kaon mass would bring to a solution to the “kaon mass puzzle”, i.e the fact that the two most precise measurements, each one with a precision of about few keV are 60 keV far apart [21].

Recently, the development of the new TES (Transition Edge Sensor, [12]) detectors, capable of few eV resolution measurements, opened a new perspective for such kind of experiments; however, two main limitations of these devices make them not easy to be used. First of all, their reduced dimensions (few hundreds of microns per side) imply the necessity to use them in arrays of some hundreds of unit in order to cover higher solid angles, having as a direct consequence a big geometrical efficiency loss due to the large dead area between adjacent sensors and, in the end, a still very small acceptance. In second place, the working principle of the TES detectors requires to reach operational temperatures below 1 K, making them very hard to handle and letting their cost explode.

Thus, a new method for the precise determination of the X-ray lines of interest, avoiding the complexity of the cooling system, but still allowing a few eV precision, is **badly needed. This is exactly what we plan to do in VOXES.**

3) Objectives of the proposal

The main objective of the VOXES project is to develop and test the first prototype of a **high resolution and high efficiency X-ray spectrometer** in the range of energies from 2 keV to 20 keV, using HAPG bent crystals in the von Hamos configuration coupled to CCD and SDD detectors for **extended sources**.

This objective will be achieved through an optimisation of the curvature radius (R), the angular aperture (θ), the thickness, the mosaicity and the vertical dispersion Δy of the HAPG crystals in collaboration with one of the major HAPG producers (OPTIGRAPH,[1]).

The VOXES system will have:

- Capability to measure X rays in a broad energy range (from about 2 to 20 keV);
- Energy resolution level of $\Delta E/E < 10^{-3}$ (meaning few eV);
- Capacity to work with extended X-ray sources;
- Reasonable cost.

Other objectives of the project are:

- To perform R&D activity together with the OPTIGRAPH company for the development of HAPG crystals and deliver versatile geometries able to work in various environments;
- To realise a competitive measurement of the pionic helium at the PSI laboratory and implement a strategy for the kaonic helium and kaon mass measurements at DAΦNE or at JPARC.
- To organise a dedicated workshop at the end of the project, where to invite representatives of the potentially interested companies/industries and other interested users.

4) Description of the proposed research activity underlying research units description and duties

The goal of the VOXES project is the construction of a **high resolution and efficiency X-ray detection** system for extended sources, based on CCD and SDD detectors and on the use of Highly Annealed Pyrolytic Graphite crystals (HAPG) in a flexible geometry.

The research activity aims to the construction of an X-ray spectrometer, consisting in three von Hamos spectrometers (VH units); each spectrometer, optimised for an energy range, $E_1 < E < E_2$, will be positioned on a dedicated spot on the Rowland circle (see Fig. 4). A flexible circular rail will allow to adjust the positions of the VH units in order to cover the energy range of interest for a specific measurement.

The following institutes are participating to the VOXES project, having the duties stated below:

- **LNF-INFN, Coordinating Unit**, will coordinate the whole project, and will be directly involved in all phases of the project; it will be responsible for DAQ and Slow Control, for the detector assembly (first and second prototype) for the tests measurements in laboratory and on beam;
- **SMI-Vienna** is responsible for the realisation of the cryogenic and vacuum systems, and will collaborate to the mechanics design and realisation, to detector system assembly and tests;

LNF-INFN unit will collaborate with **OPTIGRAPH** company to optimise and realise the HAPG crystals, being responsible for this operation and for the data analysis and with **Milan Polytechnic** for the readout electronics.

The *first phase* of the research activity during the *first year* of the project will be devoted to the single VH unit design, construction and tests, aiming to characterise and optimise the system, and to obtain the largest possible acceptance (source size) and energy resolution, which will have an impact on the final prototype spectrometer and on the target cell dimensions and detectors arrangement. Detailed Monte Carlo simulations will be realised and the HAPG crystals composition and geometry optimised; the target cell in this phase will be equipped with an adjustable slits system defining the X-ray source dimension. In this phase SDD and CCD detectors will be used. The system will be characterised in the laboratory, using targets activated by an X-ray tube. This part will be done in collaboration with the OPTIGRAPH GmbH team, SMI-Vienna and Milan Polytechnic.

In the *second phase* of the project, to be realised in the *second year*, the construction and test of the final VOXES system prototype will be realised. The mechanical part and the detector geometry assembly will be performed by LNF in collaboration with Milan Polytechnic, respectively. The cryogenic and vacuum systems will be realised at the SMI. The system will be assembled in the LNF-INFN laboratory where a first set of measurement will be done in order to obtain the proper calibration

and test all the SlowControl and DAQ software. Finally, the measurement of the pionic helium X-ray lines will be performed on the π -M1 line at the PSI and it will represent the final goal of the whole project.

The VOXES final prototype will consist in 3 VH units; the first will be optimised for energies around 8 KeV, to be used for π He, while the other 2 VH units will be optimised for the calibration lines at 4.5 (Ti) and 15 keV (Zr). A schematic diagram of the setup is shown, not in scale, in Fig. 4. Signal X rays (red) impinge on the ad-hoc shaped HAPG crystal (green) in the von Hamos configuration; the reflected X rays (violet) are then collected by the CCD detectors (yellow). Each arm of the spectrometer can move on a circular rail to collect the different Bragg angles (energies) and each HAPG-CCD unit can rotate on its central axis to optimise the X-ray collection and focusing.

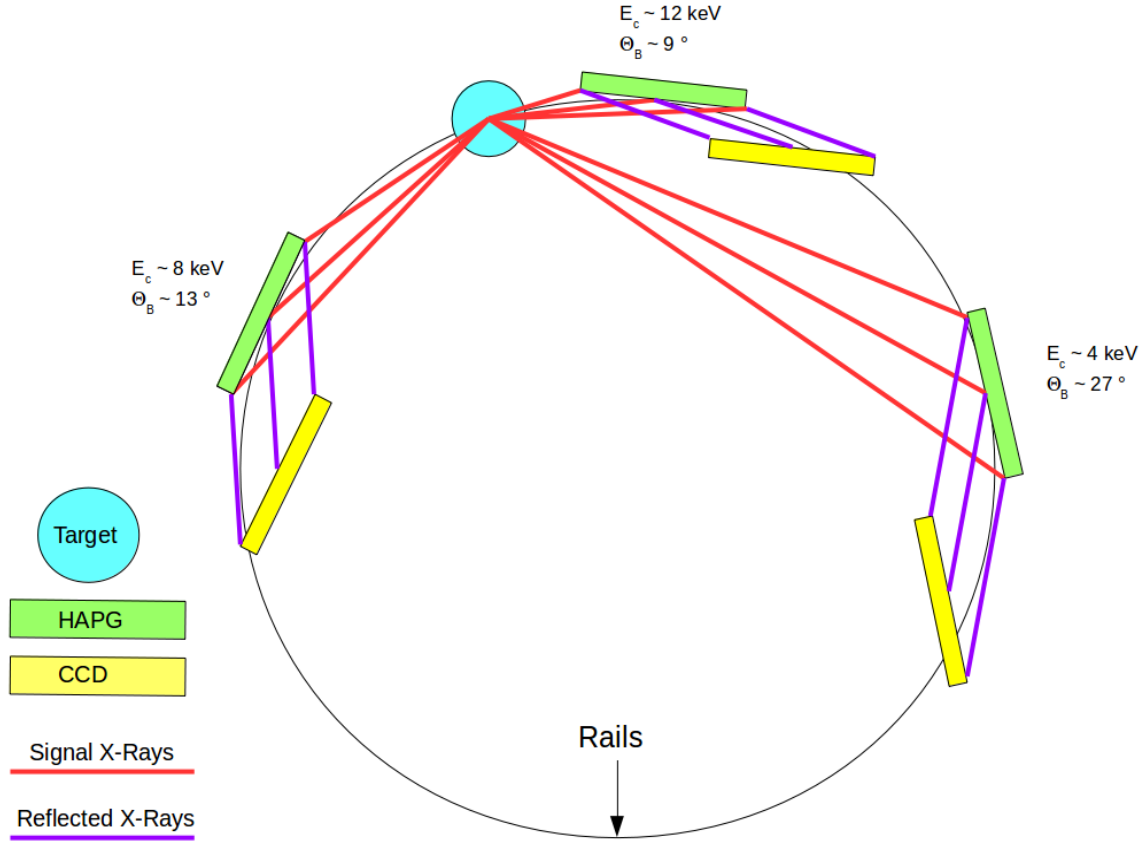


Figure 4: Schematic VOXES system prototype with 3 VH units (not in scale). Signal X rays (red) impinge on the ad-hoc shaped HAPG crystal (green) in the von Hamos configuration; the reflected X rays (violet) are then collected by the CCD detectors (yellow). Each arm of the spectrometer can move on a circular rail to collect at different Bragg angles and each HAPG-CCD unit can rotate on its central axis to optimise X-ray collection and focusing.

By the end of the second year of the project we shall, as well, organise a Workshop dedicated to high resolution X-ray measurement systems, where to present the achieved results and to invite potential interested industries and users. We shall as well prepare a proposal of use of the system for high precision exotic atoms measurements at DAΦNE and at J-PARC.

During the project an intensive dissemination activity will be realised both towards the scientific community (presentations at workshops and conferences) and towards the general public (stages, seminars and public conferences).

A schematic representation of the different phases of the project is shown in Fig. 5

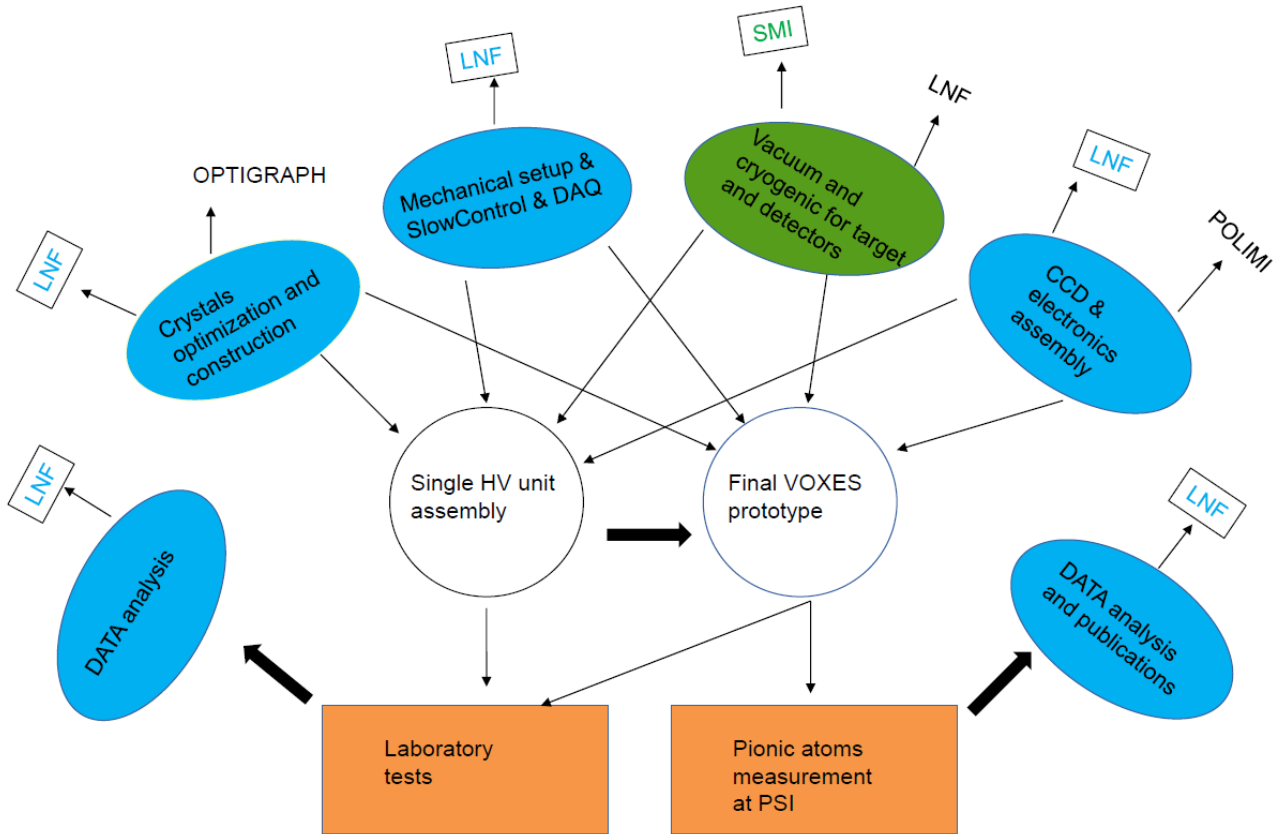


Figure 5: Diagram of the VOXES realisation phases where (in boxes) the responsible institute/unit is shown, together with all participating units.

Comment on the existing resources:

CCD and SDD detectors, as well as part of the cooling and of the DAQ system are already available within the involved institutions. The resources represents about 50% of the overall cost of the project.

5) Milestones and timeline

The VOXES project is organised in the following Work Packages:

- **WP1:** MC simulations for the parameter optimisation of the HAPG crystals and HAPG crystals realisation:

The optimisation of the HAPG crystals structure, shape, curvature radius, thickness and dimensions and their realisation will be performed in collaboration between the LNF-INFN and the OPTIGRAPH GmbH team, under the responsibility of the last.

- **WP2:** Design and realisation of the mechanical setup for the HV unit:

The mechanical setup for the single HV unit and for the final VOXES setup will be designed and realised by the LNF-INFN unit.

- **WP3:** Realisation of the cryogenic target and of the SDD and CCD detectors cooling and vacuum systems:

The cryogenic (helium) target for the test of the VOXES setup at the PSI beam, the cooling and vacuum system for the single HV units and final VOXES setup will be done by the Stefan Meier Institute (SMI) in Vienna in collaboration with the LNF group.

- **WP4:** *SDD and CCD assembly with readout electronics:*

The SDD and CCD assembly and the development of the readout electronics, for single HV unit and final setup, will be done by the LNF team in collaboration with the Milan Polytechnic.

- **WP5:** *Slow Control and DAQ system:*

Slow Control and DAQ systems will be designed and realised by the LNF-INFN unit.

- **WP6:** *First prototype and final VOXES setup assembly and test:*

The assembly of the single HV unit prototype and of the final VOXES setup will be realised in the laboratory at LNF-INFN by all the participating institutions; the systems will be characterised in the laboratory and the pionic helium measurement will be realised at the PSI – Villigen

List of major milestones (M) and deliverables (D):

- **M1:** *Laboratory characterisation of the single HV unit with X-ray tube activated targets/calibration foils:*

A laboratory test will be performed at LNF-INFN, using an X-ray tube impinging on several materials (Ti, Cu, Zr, Al, ecc..) in order to test the system response at different energies.

- **M2:** *Laboratory characterisation of the final VOXES prototype:*

A laboratory test will be performed at LNF-INFN, using an X-ray tube impinging on several materials (Ti, Cu, Zr, Al, ecc..) in order to test the VOXES system response at different energies.

- **D1.1:** *Single HAPG crystal production*

The HAPG crystal for the single HV unit prototype will be delivered by the OPTIGRAPH GmbH company.

- **D1.2:** *Final HAPG crystals production*

The HAPG crystals for the final prototype will be delivered by the OPTIGRAPH GmbH company.

- **D2.1:** *Construction of the single HV unit mechanical setup*

The mechanical setup construction for the single HV unit will be realised.

- **D2.2:** *Construction of the final VOXES mechanical setup*

The final mechanical setup, consisting in 3 HV units and the moving mechanism, will be realised.

- **D3.1:** *Realisation of the vacuum and cryogenics for the single HV unit prototype*

The realisation of the vacuum and cryogenics for the single HV unit will be finalised.

- **D3.2:** *Realisation of the VOXES target cell, vacuum and cryogenics*

The realisation of the VOXES target cell, vacuum and cryogenics will be finalised at SMI, in collaboration with the LNF group.

- **D4.1:** *Detector and electronics for the single HV unit*

The detector assembly with the readout electronics for a single HV unit will be finalised..

- **D4.2:** *Detector and electronics for the final VOXES setup*

The detector assembly with the readout electronics for the final VOXES setup will be finalised.

- **D5:** *Report on the measurement of pionic helium*

The π He measurements will be performed on the π M-1 line of the Paul Scherrer Institute (PSI) of Villigen, in order to characterise the system performance in a real measurement; the aim is to achieve an eV precision measurement and a resolution of the VOXES setup at the level of few eV. Results on pionic helium measurements will be reported and sent for publication in a peer-reviewed journal.

• **D6: Results publication**

The results of the VOXES project will be sent for publication in a peer-reviewed journal.

Month	1-3	4-6	7-9	10-12	13-15	16-18	19-21	21-22	22-24
WP1: MC simulations for the parameter optimisation of the HAPG crystals and their realisation		D1.1					D1.2		
WP2: Design and realisation of the mechanical setup for the HV unit			D2.1				D2.2		
WP3: Realisation of the cryogenic target and of the SDD and CCD detectors cooling and vacuum systems			D3.1				D3.2		
WP4: SDD and CCD assembly with readout electronics			D4.1				D4.2		
WP5: Slow Control and DAQ system									
WP6: First prototype and final VOXES setup assembly and test					M1			M2	D5
Workshop organisat.									
Dissemination (publications)									D6

Table 1: Work Packages, Deliverables and Milestones table

6) Impact (scientific, technological, socioeconomic)

The goal of the VOXES project is the construction of a high resolution and high efficiency von Hamos X-ray Spectrometer for the measurement of extended sources, using Highly Annealed Pyrolytic Graphite (HAPG) crystals combined with Charged Coupled Devices (CCDs) and/or Silicon Drift Detectors (SDDs). The innovative VOXES system will have important applications in science and technology, and could have also a strong socioeconomic impact.

• **Scientific impact**

There are many sectors in science where the determination of the X-ray energy with a resolution at the level of eV is mandatory. VOXES provides the prototype of such a system.

As an example, in hadronic physics VOXES could be used to perform fundamental measurements of the X-ray energies in the range from 2 to 20 keV with a precision of few eV and with a width of the lines as small as few eV; this system can then be used to perform the world's most precise kaon mass determination[18], or the isospin-dependent part of the strong interaction in the low energy strangeness sector from the $K^4\text{He}$ and $K^3\text{He}$ atoms X rays [2,3,4]. Such measurements were not yet done due to the intrinsic limitations coming from the solid state devices used up to now as X-ray detectors.

Other sectors where VOXES can be used are: test of the validity of the Pauli Exclusion Principle by measurements of the prohibited X-ray transitions in atomic systems or measurements performed using the synchrotron radiation.

- *Technological impact*

The VOXES project will represent a breakthrough in the field of X-ray optics; presently, all mirrors and experiments are usually carried out using very small size and angle sources to exploit at best the focusing properties of such systems. The results obtained by VOXES will prove the possibility to use the extremely good focusing properties of the bent HAPG films for a complete new set of experiments where the signals coming from non-micrometric sources could be collected and measured. For example, the current generation of direct-viewing astrophysics telescopes is still nowadays heavily noise limited. It can accurately study only the strongest sources. Thus focusing of hard X rays is mandatory in order to overcome these sensitivity limitations [22]. In this field, VOXES could represent a breakthrough allowing a higher collection for wider aperture angles of the incoming radiations. In synchrotron radiation, VOXES could as well represent an improvement in the beam monitoring and focusing with respect to nowadays techniques [23].

Moreover, the possibility to use the VOXES-type systems at the X-FEL facilities for various type of measurements represents an opportunity to extend the area of application of such systems.

- *Social impact*

In medical physics, mosaic crystals are widely used, for example, in mammography [24, 25]; the proposed system would allow to reduce the X-ray dose thanks to a higher collection efficiency.

The VOXES project could also be used for high precision elemental mapping of different materials, contributing in several fields like geology, archeometry and chemistry [26].

In Fig. 6 there is schematically shown the potential impact VOXES has in science, technology and for society.

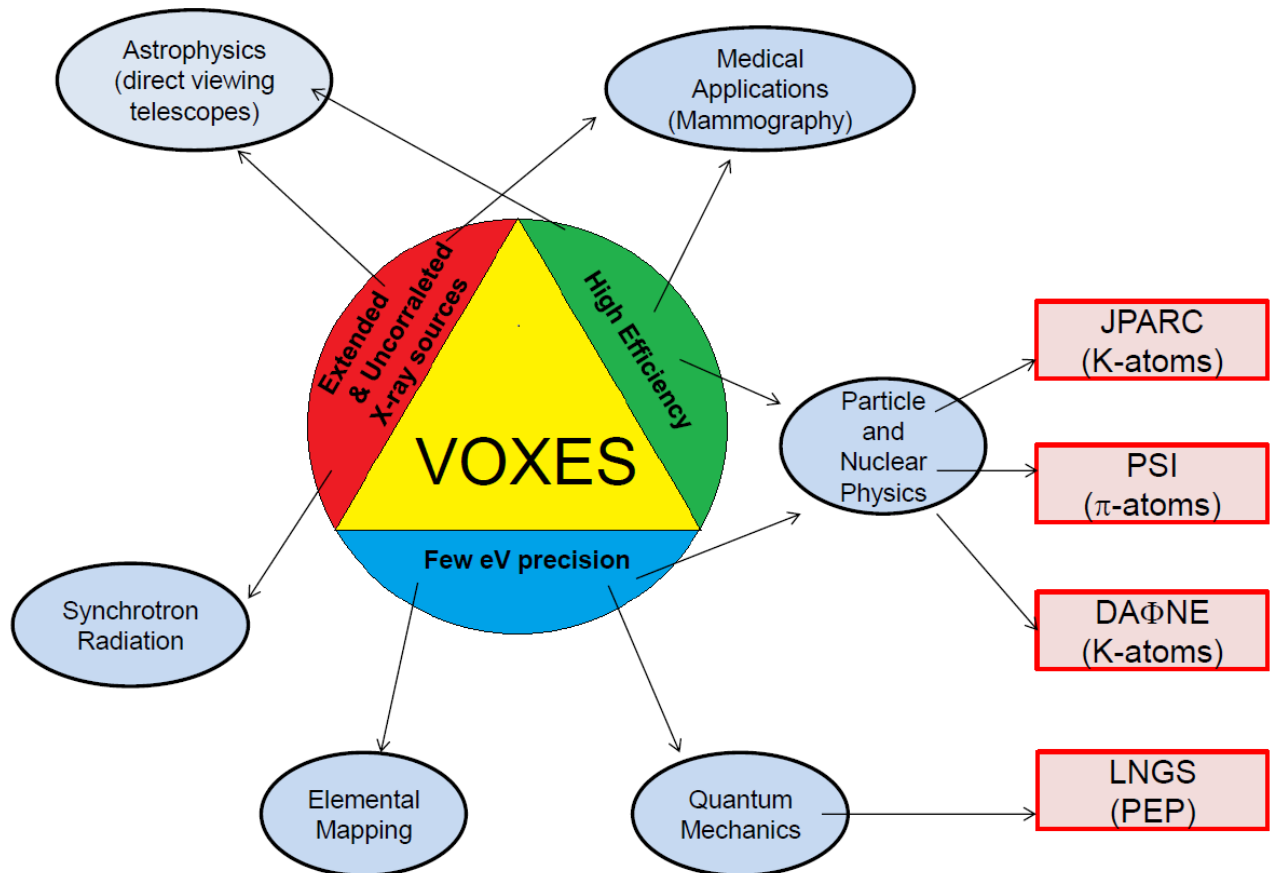


Fig 6: Impact of the VOXES system in science, technology and society.

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